

# Strategic Reasoning and Assumption Testing: The Cognitive Foundations of Strategy

## Executive Summary

Strategy is only as good as its underlying assumptions. Under Active Inference, strategic reasoning is the process of building, testing, and updating the generative model the organization uses to predict competitive outcomes. This module examines strategic assumption analysis, cognitive biases in strategy, scenario thinking as hypothesis testing, and the discipline of making implicit beliefs explicit and testable.

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## Learning Objectives

1. Define **strategic reasoning** as inference about competitive dynamics under uncertainty
  2. Apply **assumption-based planning** to surface and test strategic beliefs
  3. Identify **cognitive biases** that distort strategic reasoning
  4. Use **scenario thinking** as a method for evaluating competing strategic models
  5. Design processes for **systematic assumption testing** in strategic contexts
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## Key Concepts

### 1. Strategic Assumptions

Every strategy rests on assumptions about:

- **Customer needs:** What do customers value? How will preferences evolve?
- **Competitive dynamics:** How will competitors respond? Who will enter/exit?
- **Technology trajectory:** What will be possible in 3, 5, 10 years?
- **Regulatory environment:** What rules will govern the industry?
- **Macroeconomic conditions:** What will the economic environment look like?

**Case Study — Intel's Strategic Inflection Point:** Andy Grove described "strategic inflection points" as moments when the assumptions underlying a company's strategy are invalidated. Intel's shift from memory to microprocessors in the 1980s required recognizing that a core assumption (Intel is a memory company) was no longer viable. The cognitive challenge was enormous — the assumption was so deeply embedded that the company initially couldn't see the change.

## 2. Cognitive Biases in Strategy

| Bias                     | Strategic Impact                                                                      | Mitigation                                            |
|--------------------------|---------------------------------------------------------------------------------------|-------------------------------------------------------|
| <b>Sunk cost fallacy</b> | Continuing failed strategies because of past investment                               | Evaluate options based on future value, not past cost |
| <b>Planning fallacy</b>  | Systematically underestimating time, cost, and risk                                   | Use reference class forecasting                       |
| <b>Overconfidence</b>    | Overstating the precision of strategic predictions                                    | Calibrate predictions against historical accuracy     |
| <b>Status quo bias</b>   | Preferring current strategy even when alternative is better                           | Assign advocates for change options                   |
| <b>Narrative fallacy</b> | Constructing coherent stories to explain outcomes that were actually driven by chance | Demand causal mechanisms, not just stories            |

## 3. Assumption-Based Planning

A structured process for surfacing and managing strategic assumptions:

1. **Identify** assumptions underlying the strategy
2. **Classify** by importance and vulnerability
3. **Define** signposts — observable indicators that an assumption may be failing
4. **Design** shaping actions (to make vulnerable assumptions more likely to hold)
5. **Prepare** hedging actions (contingency plans if assumptions fail)

## 4. Red Team / Blue Team Analysis

Deliberately stress-testing strategy by assigning a team to find weaknesses:

- **Blue Team:** Defends the current strategy, articulates its logic
  - **Red Team:** Attacks the strategy, identifies assumptions most likely to fail and scenarios under which the strategy collapses
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## Cross-References

- For organizational cognition, see [Organizational Systems: Cognition](#)
  - For team sensemaking, see [Collective Intelligence: Cognition](#)
  - For algorithmic decision support, see [Digital Transformation: Cognition](#)
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## Summary

| Concept                   | Strategic Modeling Meaning                                        |
|---------------------------|-------------------------------------------------------------------|
| Strategic reasoning       | Inference about competitive dynamics under uncertainty            |
| Strategic assumptions     | Implicit beliefs underlying every strategic plan                  |
| Cognitive biases          | Systematic distortions in strategic reasoning                     |
| Assumption-based planning | Surfacing, classifying, and monitoring strategic assumptions      |
| Red Team analysis         | Deliberately stress-testing strategy through adversarial critique |

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## References

- Grove, A. S. (1996). *Only the Paranoid Survive*. Doubleday.
- Dewar, J. A. (2002). *Assumption-Based Planning*. Cambridge University Press.